Freight rate prediction

Approach and results | Musaab

I used the 48,000 labeled loads to predict the total price of each shipment in the 12,000-row final file. I started with a price-per-mile baseline, then compared it with one gradient-boosting configuration. After checking the results on later dates, I refitted the selected pipeline on all labeled data.

How I split and checked the data

The labeled data runs from January to October 2025, while the final inputs cover November and December. I split by date so that each evaluation used shipments later than the training period.

Stage	Training period	Evaluation period
First comparison	Jan-Jun: 28,806 loads	Jul-Aug: 9,671 loads
Later-date check	Jan-Aug: 38,477 loads	Sep-Oct: 9,523 loads
Final fit	Jan-Oct: 48,000 loads	Nov-Dec: 12,000 predictions

I fitted preprocessing separately for each training period and kept the model settings fixed for the later-date check. My initial exploration included the full labeled file, so I treat these scores as local validation estimates. Spotter holds the final labels.

Results

Evaluation	Approach	MAE ($)	RMSE ($)
Jul-Aug	Price-per-mile baseline	249.01	674.72
Jul-Aug	Gradient boosting	160.27	627.70
Sep-Oct	Price-per-mile baseline	256.95	684.25
Sep-Oct	Gradient boosting	110.02	634.82

The model reduced MAE by 35.6% in July-August and 57.2% in September-October against the corresponding baseline. September and October MAE were $106.82 and $113.09. RMSE stayed much higher because a small number of loads had large errors.

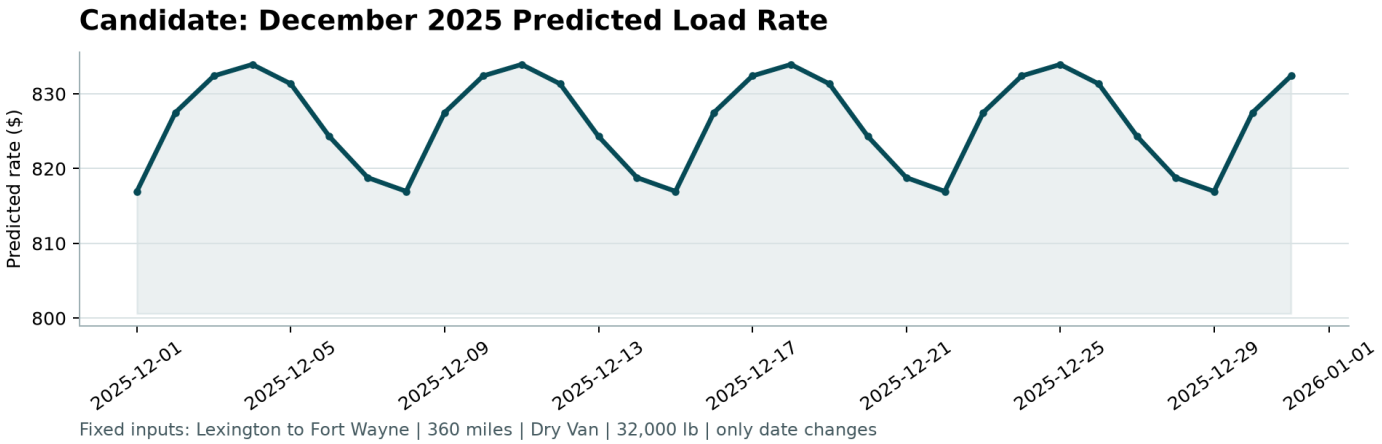
What went into the model

I used pickup, delivery, distance, equipment, weight, and date, since these fields are available for both the main predictions and the December chart. I represented date with month and weekday. Nonpositive weights became missing values; the pipeline filled them using training medians and recorded missing-value indicators. Cities and equipment were one-hot encoded, with unfamiliar categories handled by the encoder.

I used HistGradientBoostingRegressor with absolute-error loss to focus on average dollar error. Settings were learning rate 0.05, 300 iterations, 31 maximum leaves, 30 minimum samples per leaf, L2 regularization 1.0, early stopping disabled, and seed 42. The baseline was distance multiplied by the training median price per mile.

December predictions and remaining limits

Fixed load: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb



The predicted price ranges from $816.96 to $833.93. Only the date changes in this example. Since all rows are in December, the variation comes from weekday, which explains the repeating weekly shape. The chart was generated by the supplied score.py.

Data issues and large errors

Issue	Labeled data	Final inputs
Missing weight	300	165
Nonpositive weight	292	145
Missing market_index	374	249

I retained all evaluation rows. For example, a Phoenix-to-Syracuse load had a recorded price of $24,294.98, while the model predicted $5,340.84. I could not verify that the recorded price was incorrect, so I kept it in the error calculation. This kind of miss helps explain the gap between MAE and RMSE.

What this version leaves out

I left coordinates, market_index, and quote_signal out of this version because the December inputs do not contain them. Using the shared fields keeps one consistent prediction path, but gives up information that could improve the main predictions. Missing market_index values therefore do not affect this model.

November and December are outside the observed training months, and the model has no external market or holiday inputs. Its December pattern needs validation against actual prices. I would next compare a richer model for the main dataset while keeping an explicit approach for the chart's smaller input set.

The final outputs contain 12,000 load-ID-matched predictions and 31 completed December rows. The supplied checker accepted both files. The repository includes the code, pinned dependencies, run instructions, and this chart.